



Mariánské Lázně

*is known as a town of mineral waters
and it is famous for its unique colonnade.*

*But do we really know mineral springs
and do we know why they are here and what
they are useful for? Do we know which
natural wealth is hidden in them?*

*This is what this book is about and a great
thanks belong to their authors, who processed
this unique issue again after fifty long years.*

*Emerge into the secrets of nature and discover
the magic of mineral waters.*



Short history of mineral springs in Mariánské Lázně

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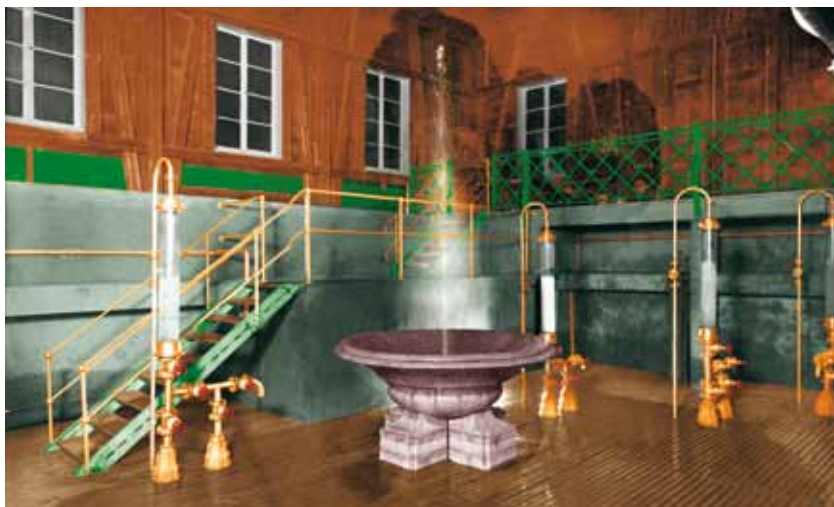
Cross spring (Křížový pramen)

Cross spring is one of the longest used springs in Mariánské Lázně. The first written documents about its use for a mineral water drinking cure, when it served as a laxative, originate from the beginning of the 17th century.



In 1749, the Teplá pharmacist Damián Schulz started evaporating water from Cross spring and the laxative salt that he gained, he then sold it in a wide area under the name 'Sal Teplensis'. He had a wooden cross erected near the spring, from which the spring got its name. At this time, the first documented alteration of the spring was carried out. The surroundings of the spring were cleaned, the surface water was conducted away from the spring and flat stones were laid around an outflow itself. In 1790, the monastic doctor, Jan J. Nehr, carried out the first proper capture of the spring into a wooden octagonal reservoir inset with a ring of polished serpentinite. At that time, one more mineral spring, called Gagging spring (Brechsäuerling) arose in the vicinity of Cross spring. It had similar effects as Cross spring, but whereas it was not properly captured, it was polluted and less used by people. In 1818, Gagging spring was filled in and a new representative pavilion with its Ionic columns was erected above Cross spring. The capturing itself has been altered several times up to today, due to the fluctuation of intensity, which was the result of blockages of the outflow paths. Probably the most extensive alterations were done during 1911-1913, when the original pavilion was taken down and the earth was excavated to a depth of 4 m below the terrain. There were over 36 outflows captured by tin bells in an area of 200 m² and the water was collected in a collecting tank from where it was led into a pour-over vase. Even this alteration was not the last and in 1936 the intensity was secured by wells. In total, 3 wells were made - Cross spring I captured at a depth of 17.4 m, Cross spring II at a depth of 30 m and Cross spring III at a depth of 45 m. In 1945, Cross spring I totally disappeared, the intensity of Cross spring II decreased to 0.29 l/min. Cross spring reached its highest intensity, 3.3 l/min, in 1953. Then in 1955 it rapidly decreased and it started to spout interruptedly. During 1955-56, reconstruction of the wells was carried out. Well No. I and II were sealed up and the original well No. III, under the dome of the pavilion was widened, cleaned and newly equipped. A new well No. IV was drilled in front of the J. J. Nehr bust, with a wide outflowing profile and at the depth 83.5 m, which captured water

with a higher amount of solid particles than well No. III. This last alteration continues to serve without changes up to today.



Maria spring (Mariin pramen)

Maria spring is probably the longest used medicinal resource in Mariánské Lázně.

The information comes from 1609, when Jáchym Libštejský from Kolowraty was treated here with water from the so called Stinking spring (Stinker), following advice from doctor M. Raudenius from Horní Slavkov. He probably had locomotor problems and his treatment was apparently successful. The name of the spring is based on a legend, when one war veteran cured his lame legs here and as a gratitude to the Virgin Mary for his healing, he put her picture on a nearby tree. The picture then served as a sign for everybody, who was looking for the spring and so the spring began to be called Maria spring. The first mention of this name dates back to 1766. Outflows of carbon dioxide together with outflows of mineral water were a part of a smaller peat bog, which was, at the beginning of Mariánské Lázně, used as a source of peat for peat com-



presses and wraps. At first the outflows were fenced only by wooden logs, so they created a small pool with bubbling mineral water. Later they were roofed so the guests would be sheltered from bad weather. The springs underwent their most extended balneologic alterations during 1929-1932. A shaft was dug out of an area of 22 x 6 m, where fragmented outflows of mineral water were

found. All together 6 outflows, Maria No. I – VI, were captured, from which five were fitted with separating cylinders for obtaining CO₂ gas. Maria III did not have a separating cylinder and it was conducted into the spring vase, where it resembled the spurting Hot spring from Karlovy Vary (Vřídlo). In an adjoining area, a pool was built, filled with normal water, which seeped through and mixed with underground mineral water and by hydrostatic pressure pushed the mineral water into the separating cylinders. At the end of the 1980's, this pool was filled in and the water sank under floor level. The area under the gallery was filled with carbon dioxide. The last alterations were done in 2006, after damage to the old roof construction and from the technical area it became a gas pool, from which the gas is pumped for treatments up until today.

Ambrose springs (Ambrožovy prameny)

Ambrose spring, named in honour of Teplá abbot Jeroným Ambrož (Ambrose), was described by A. J. Scrinici in 1760 as a mineral spring used for general drinking.

In 1766, J. J. Zauschner described it in his work under its present name. Correct capturing into wooden boarding, common at that time, was done in 1807 and it lasted in the same place until 1911, when it was replaced by capturing into stoneware pipes. In 1826, the original simple wooden gazebo was replaced by a neo-Gothic pavilion. Regarding the fact that the iron content, due to which the spring was used for mineral water drinking treatments, rapidly decreased in 1925. During the next year, a total reconstruction of its capturing was done. Three groups of outflows were captured by conically built well from burnt bricks at a depth of 7m under the terrain and they were led by three pipes into outpouring vases. This way of capturing has remained unchanged until the present day. The springs vary little in their chemical composition.





Ferdinand spring (Ferdinandův pramen)



Ferdinand spring is the oldest recorded spring in Mariánské Lázně.

It is mentioned in a letter from king Ferdinand I to a Teplá abbot, where the king asks for samples of the spring water to be sent for analysis, if it could be the source of kitchen salt, which was at that time missing in the

Czech kingdom. In historical documents the spring is always mentioned in connection with experiments to mine salt by evaporation. The outflow for many years looked like a wet swamp with efflorescence of mineral salts. The first proper capturing was done in 1819 by a wooden shaft at a depth of 6.5m. In that year it received its present name after the Czech king Ferdinand I, until then it was called Úšovický spring. In 1869, water from Ferdinand spring started to be piped to Nové Lázně and one year later to the main colonnade. Piping caused the overall fluctuation in the chemical composition of the spring and its mixing with normal water. During 1922-26, an extended survey was carried out, which was finished in 1930 by the capturing of 8 new outflows from the Ferdinand spring area. There are two mineral springs of various chemical composition captured directly under the central dome. Ferdinand I is captured by a well at a depth of 20.5m. While drilling the well, mineral water was discovered at a depth of 5m, later captured by four cone-shaped units and named Ferdinand III. Ferdinand IV differs in its chemical composition from the other waters from the Ferdinand springs group, due to its low concentration of minerals and a high content of carbon dioxide. Mineral water from this source received the name Excelsior and it was once bottled and sold under this name.





Caroline spring (Karolinin pramen)

The first records date from 1809, when the spring was discovered during alterations to a promenade area between Cross spring and Central Spa (Centrální Lázně).

As it was a newly discovered spring, it was simply called New spring. Its present name was given in 1819 in honour of then Empress Caroline Augusta. The first simple gazebo with eight Corinthian columns was built over the spring in 1811 and it was slightly altered in 1823. It remained in the same condition until 1870, when Ferdinand spring was brought from Úšovice to its place below a church. The original gazebo was pulled down in 1872 and replaced by a raised two-wing colonnade. During reconstruction of the colonnade, the capturing of the spring was renewed at a depth of 5.7 m. Caroline spring was not led to the colonnade but it was piped to Nové Lázně for making baths. In 1929, due to the decreasing spring intensity, a new capturing was done at a depth of 11 m, but the spring was still used only for baths until 1981, when it was brought back to the colonnade. During 1986-90, a complete reconstruction of the colonnade was carried out, the old colonnade from 1872 was demolished and its replica was built 12.8 m nearer to Cross spring. The original outflow of Caroline spring is symbolically depicted on the stone floor.

Forest spring (Lesní pramen)

Forest spring was very likely used at the end of the 17th century. The first recorded capturing and a simple gazebo with a thatched roof comes from 1827-28.

In the same year, professor Steinmann from Prague performed the first analysis. In 1840, an old shelter was replaced with a new octagonal gazebo carried by two rows of columns. Due to the bad condition of capturing from 1827, a new one was carried



out in 1869. At the same time an old pavilion was pulled down and a new colonnade, according to the plan by Mariánské Lázně architect Friedrich Zickler, was built. The last reconstruction is from 1905. The spring has a bell capturing at a depth of 6.9 m under the terrain. The original open and walk-through pavilion was enclosed in glass in 1956. In 1999, during reconstruction of the sewerage system, the spring was contaminated by styrene, toluene and other dangerous chemicals. For that reason, it was prohibited to use the spring and a new suitable source of similar mineral water was sought in the vicinity. Today the spring is harmless and suitable for general use.

Balbin spring (Balbínův pramen)



Balbin spring was discovered in 1853 in a peat deposit, which had started to be used at that time in spas.

While clearing peat, three springs were gradually discovered and they were captured in simple shallow wooden tanks. Simple shelters for spa guests were built above the springs. The springs were never used for mineral water drinking treatments and they served only as a trip destination. It changed in 1921, when they were captured into perforated stone pipes and drainage. The spring water was collected by piping into Nové Lázně, where it is still used today for carbonated baths. Stoneware pipes were replaced by a well

and a simple overflow with a drinking fountain was established in the place. Today the water is piped and accessible by the statues of Monarchs directly in Mariánské Lázně.



Rudolph spring (Rudolfův pramen)

Near the present Rudolph spring gazebo, the so called Meadow spring (Luční pramen) used to arise on the meadow, which belonged to Úšovice farmers. Teplá monastery strived to buy it for many years.



It was analysed for the first time in 1820 and it was recommended for use. However, farmers did not want to give up their land, so the spring stayed for a long time without any alterations.

The monastery succeeded in purchasing a neighbouring plot in 1865, and then preparatory work on capturing the spring started. The original Meadow spring was conducted away into a shallow side reservoir and analysed the next year and in summer 1867 it was named 'Kronprinz-Rudolfsquelle' – Crown Prince Rudolph spring, in short Rudolph spring. A simple glassed gazebo was built above the spring. In 1902, reconstruction of the capturing, which was deepened to 4 m below the terrain, as well as the whole gazebo. The original shallow capturing could not prevent the spring mixing with ordinary underground water in the meadow around a stream, therefore in 1937, a 10 m well was drilled through the centre of outflowing pipe. But even this alteration did not stop the fluctuation of mineralisation of Rudolph spring. This outflow is called Old Rudolph. Beside it, there are several captured springs in the vicinity – in 1902 Rudolph I and Rudolph II, in 1910 Rudolph III. The presently bottled mineral water called Rudolph spring comes from the wells BJ-36 and BJ-37 (Rudolph VI).

Alexandra spring (Alexandřin pramen)



In 1870, a new spring was discovered by the spa company (Curorte-Baugesellschaft) at the heel of a slope under today's Main street, it was temporarily called Gisela spring.

In 1872, a capturing was done correctly and in 1873 it received a new name Alexandra spring after her majesty Alexandrine of Prussia, Grand Duchess of Mecklenburg-Schwerin. During the reconstruction of the spring in 1900, an outflow of another spring was discovered and it was named after the recently deceased abbot - Alfred Clementso, Alfred spring. Both springs were captured by cones at a depth of 5 m in the same year. In 1938, the intensity decreased to just 0.83 l/min and in 1941, after the reconstruction of the spring, it increased to 4 l/min. Since then its intensity continually dropped until 1953, when it was nearly unmeasurable. The gazebo was demolished in 1957 and the spring was not used for many years. In 1987 the spring was newly captured at a depth of 60 m and today it is used in hotel Falkensteiner Grand MedSpa Marienbad.

Antonin spring (Antonínův pramen)

Antonin spring was known to citizens of Úšovice for a long time. According to descriptions and the old location, it is hard to tell, if they are referring to the present Antonin spring or the nearby Prelat spring. It is certain that in 1933, the spring was captured near the railway embankment and the water was piped into the drinking vase in Úšovice village square. The spring was not used as curing water, but only as common refreshment water, to which it serves up to today. In 1986, a new modern roofing, by the architect M. Břixa, was built above the original spring vase from 1933 and also the piping of mineral water was altered.

Joseph spring (Josefův pramen)



One of newer springs in Mariánské Lázně is Joseph spring. It was captured in 2001 by a 100m deep well HV-40 and it gained the status of a natural resource in 2003. It was drilled during a period, when looking for a new natural medical resource which would replace Forest spring. It is situated in a watery meadow near Forest spring, it has half the amount of minerals than other springs Mariánské Lázně, but it has the highest content of free carbon dioxide. It is piped into the nearby spa centre Palladio, where it is used for baths and it can be

also used for a mineral water drinking cure. The spring was named after the original owner of hotel Palladio, architect Josef Schafer, according to whose plans the important spa buildings in Mariánské Lázně were built. These were for example - hotel Nové Lázně, Cultural and Conference Centre Casino, the building of the present secondary grammar school (Gymnázium a Obchodní akademie) and also the very building of villa Palladio itself.

Edward and Augustin springs (Edwardův a Augustinův pramen)



Both springs belong amongst totally new springs, piped from a depth of 76m from the Ferdinand springs area, to hotel Cristal Palace and hotel San Remo for medical treatments. Augustine spring, named after the chairman of the board of the company Cristal Palace, a. s., Alfred Augustin, is captured by the well HV-43. Edward VII spring is captured by the well HV-44 and it is named after the English king Edward VII, who altogether visited Mariánské Lázně nine times. They were certified as natural medicinal resources in 2014. A drinking fountain for both springs was ceremonially opened and blessed during the Spa season opening, 9th May 2015 in the presence of the Minister for Regional Development, Karla Šlechtová.

Geologischer Schnitt durch die Mineralquellen von Marienbad





Mineral water drinking cures using mineral springs from Mariánské Lázně

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The term mineral water drinking treatment is understood to be a regular peroral intake of mineral water with medicinal effects by a patient with a certain illness during a limited period according to a medical prescription. A doctor chooses the most suitable mineral water, decides on its amount as well as the intake of mineral water allocation during the day.

The prescribed mineral water has to be a certified natural medicinal resource according to the Spa Law (Act No. 164/2001 Coll.) and evaluated on the basis of criteria stated in an annex of the Degree on spas and resources (Decree of the Ministry of Health 423/2001, an annex 1.a). Further, it is required that the mineral water is stable in its properties and it is harmless from a health and microbiological point of view. The mineral water drinking treatment has a proven or expected pharmacological effect and it is the pillar of spa cure of gastro-intestinal, nephron-urologic and some metabolic diseases. We should never mistake it for respecting the fluid intake, which only serves to maintain normal hydration of the organism by various drinks (the basis is drinking water and other fluids). Drinking treatment can be prescribed only to patients with balanced salt and water management and a free outflow of urine. The organism holds only the necessary amount of water and salts and the excess is discharged especially by the kidneys and the digestive system. Most often, patients undergo the drinking treatment during their spa stay together with other medical treatments, which mutually intensify their effects. An advantage is a wide choice of mineral springs of various physical properties and chemical compositions, such as in Mariánské Lázně. Another possibility is home treatment, which is limited by the number of types of mineral water with medical effects bottled and sold within the network of pharmacist's and some kinds of supermarkets. The drinking treatment is suitable for treating subacute and chronic diseases. The main contraindications of a drinking treatment are all conditions, which are accompanied by advanced insufficiency of the heart, liver and kidneys and imperforation of the urinary tract. Caution is necessary with hypertension (pay attention to a higher concentration of sodium especially combined with chlorides), patients with epilepsy, glaucoma (no large individual dosages), small children and elderly people. Skin effusion, weight increase between days and an increase of blood pressure are signs of fluid retention.

If it occurs, it is necessary to decrease the daily intake of mineral water by half or totally discontinue. For further treatment prescribe less concentrated mineral water.

The type and amount of mineral water can be chosen according to the basic illness, existence of a side illnesses (due to positive effects and contraindications) and according to the patient's body weight considering individual tolerance and age. Only water which is drunk is effective. According to Fried there is small (5 ml/kg/day), medium (10 - 15 ml/kg/day) and large (20 - 30 ml/kg/day) dosage. We divide the total daily dosage into individual amounts twice to six times a day, before, after or in between meals. Due to the high content of free carbon dioxide in Mariánské Lázně's springs, it is recommended to drink mineral water slowly, sip after sip, from spa cups while walking slowly. Spa cups for small children of the age of 3 - 6 have a volume of 100 ml, for school children 7 - 12 years 150 ml, for adults and children older than 12 years 220 ml. If more cups are prescribed in one dose, it is recommended to have a break of 10 - 15 minutes in between the cups. The effect of the drinking treatment has been proven using bicarbonate, sulfate, chloride and carbonate mineral waters containing sodium, potassium, calcium and magnesium.

What do we understand by the term medicinal mineral water?

According to the Spa Law, mineral water used for medical treatments is defined as naturally occurring underground water of original purity containing at least 1 g/l of dissolved solid matter or 1 g/l of dissolved carbon dioxide or water containing another, medically important chemical element or which has a higher temperature than 20°C at its outflow or radon radioactivity higher than 1.5 kBq/l. In Mariánské Lázně, the effects of the used types of mineral water are based on the amount of dissolved mineral substances and the amount of carbon dioxide.

We work with types of mineral water of an overall mineralisation which varies from slightly to very strongly mineralised water. 3% and stronger solutions of mineral salts



are not used for drinking treatment because they strongly irritate the mouth and stomach mucosa, causing a burning feeling and induce vomiting. The types of mineral water we use are complicated ion drinks, which have positive as well as negative effects on the organism, that means they have indications and contraindications.

No mineral nor medicinal water is suitable for permanent drinking!

Patients with kidney and urinary tract illnesses - with drinking treatments we try to influence:

1. Diuresis (increase) and frequency of urination (increase or normalise).
2. Decrease risk factors of urolithiasis
 - by increasing water diuresis and diluting urine, decrease the concentration of stone-forming substances,
 - by choosing a suitable mineral water, support the discharge of substances protecting from salt crystallization (increase discharge of magnesium and citrates).
3. With urinary tract infections, does not change pH too much (5 - 6).
4. Alkalize urine only with urate and cystin lithiasis (6,6 - 7,2).

After evaluating these requirements, chemical composition and physical properties of mineral water, the following acidulous mineral springs meet the conditions for basic natural medicinal resource for drinking treatment of nephron-urollogic illnesses – bicarbonate calcium-magnesium (Rudolph spring) and sodium-magnesium-calcium water (Caroline spring). These are hypotonic, medium to strongly mineralised types of mineral water with a low content of sodium (the best 200 mg/l). Mineral water of this composition and properties can be used for drinking treatment with most nephron-urollogic indications. Hypotonic sodium bicarbonate (Forest spring) and simple ferrous acidulous (Ambrose spring) types of mineral water have a limited use in drinking treatments.

Why hypotonic mineral water?

Hypotonicity (osmotic pressure under 710 kPa, i.e. 280 mOsm) and the low content of sodium in this water provide a 36% larger acute diuretic response, compared to the same amount of ordinary water (Kužel, VÚB Mariánské Lázně). The increase of diuresis, induced by hypotonic mineral water is conditioned by a slight growth of glomerular filtration and especially by the decrease of tubular resorption of water due to the lower secretion of antidiuretic hormone by neurohypophysis. The higher concentration of sodium, namely in hypotonic mineral water, slows down its discharge and it decreases its 'washing effect' and increases calciuria. The more osmotic pressure of hypotonic water is nearer to isotonicity, the more the water is retained and more slowly discharged. This antidiuretic effect has been proven from the concentration of sodium of 500 mg/l (7).

Influencing urine pH

If the mineral water contains bicarbonate anion bound to calcium and magnesium up to the concentration of 1500 mg/l, pH of the discharged urine does not change significantly (the strength of the bond decreases its alkalogenic effect). There is only partial dissociation and absorption of Ca_2^+ and Mg_2^+ ions from the intestine (approx. 30-40%), especially if there is free carbon dioxide. Mineral water of this composition is suitable for drinking treatment with a majority of nephron-urollogic indications including urinary tract infections.

It is different with cystin and urate urolithiasis, where we want to improve their solubility by urine alkalinisation. With these stones we prefer mineral water with bicarbonate content above 1500 mg/l bound to sodium, because from this bond it can be easily released, absorbed and the excess of the bases is discharged into the urine, which is alkalinised according to the amount of the given dose. The amount of mineral water is chosen so the urine has an optimal range of pH 6.6 - 7.2. To keep the urine pH above 6.5 it is usually necessary to have a dose of 40 - 100 mekv sodium bicarbonate divided into several individual doses (1 g sodium bicarbonate is 12 mekv). We do not carry out urine alkalinisation when there is urine infection or when precipitation of phosphates around the uric acid core can occur. We also choose mineral water with a bicarbonate content up to 600 mg/l for patients with phosphate urolithiasis, with a concern of phosphate crystallization during an alkaline reaction of the urine.

What is acidulous mineral water?

Acidulous water is any mineral water containing more than 1000 mg of dissolved carbon dioxide in 1 litre. Mineral water, whose effect is based only on the defined amount of carbon dioxide is called pure acidulous water. During drinking treatment using acidulous water, the present carbon dioxide decreases the sensitivity of the taste receptors in the mouth and so it improves the taste properties of the mineral water. Cold mineral springs drunk directly by the outflow contain more carbon dioxide.

Carbon dioxide causes blood congestion of the stomach mucosa (hyperaemia), which causes an increase of stomach fluid secretion, supports motility and improves the depleting ability of the stomach. It is quickly absorbed, it moves the respiration quotient, slightly quickens breathing frequency, so it is possible to detect it in exhaled air. At the same time, it accelerates absorption of other substances as well (calcium, magnesium, sulphonamides, alcohol, water, etc.).

In higher concentrations, it increases susceptibility of CNS. This energizing effect can be pleasant during fatigue, but it is not desirable if we are going to sleep or with small children with dominating increased irritation of CNS above the imperfect internal inhibition as well as with hyperactive children. These effects also concern artificially carbonated drinks. Quick drinking of a large amount of acidulous mineral water can cause a feeling of fullness to pressure in the stomach area, belching and tympanites and with especially sensitive individuals also nausea and vomiting. Carbon dioxide evaporates faster by heating, bubbling, stirring and free standing, which is used during a drinking treatment with patients after stomach resection or hyperacidity.

Which ions are mostly present in mineral waters?

From cations there is sodium, calcium, magnesium and potassium. Sodium is present in some springs in a large amount (Cross spring - 2700 mg/l, Ferdinand spring - 2070 mg/l, Augustin spring - 1880 mg/l, Alexandra spring - 1430 mg/l, Edward spring - 1010 mg/l) which has to be considered while prescribing a drinking treatment (we add

extra sodium above its usual amount in food). Uncontrolled drinking of mineral water with a high content of sodium can cause serious health problems especially with people with cardiovascular illnesses.





Sodium

Sodium is freely soluble in water and it absorbed well from food as well as water. In relation to health risks caused by the excessive intake of sodium in population (In the Czech Republic 7 - 10 g/day) the World Health Organisation recommended the highest limit for a daily intake of sodium as 65 mg/kg of body weight per day i.e. about 4.5g for a body weight of 70kg. Discharge of sodium by urine reflects its intake.

With a drinking treatment, we bring to the organism water as well as salts (sodium) and cause oscillation of the volume and concentration of salts (sodium) in body fluids. While taking a drinking treatment with hypotonic mineral water, water resorption causes an increase of plasmatic volume and neurohypophysis reacts through volume receptors (in heart atria, vena cava and in lung vessels) with lower production of vasopressin (antidiuretic hormone, ADH). The functioning kidneys respond by decreasing the reabsorption of water in the kidney tubules while increasing diuresis (acute diuretic response), which gradually fades in order to regain homeostasis of the inner environment and to prevent dehydration of the organism.

While taking a drinking treatment with mineral water containing a higher content of sodium, neurohypophysis gets the information through osmotic receptors to increase the secretion of ADH and water retention. A consequence of this effect is the increase of plasma volume as well as overall body water. Under this influence, the secretion of renin from juxtaglomerular cells in the kidneys decreases, production of angiotensin II decreases and adrenal cortex decreases production of aldosterone and its effect on the distal tubule and excess of sodium is discharged. With the excretion of sodium, there secondly rises excretion of water until the normal volume of plasma is restored. With the rising concentration of sodium in mineral water, a diuretic effect of the drinking treatment decreases. A decrease of physiological night diuresis occurs from the concentration of 500 mg/l of mineral water.

Calcium and magnesium

The content of calcium in mineral waters used for drinking treatment for nephro-urologic patients is usually within the region of 150-300 mg/l and magnesium above 50 mg/l. Only for patients with calcium lithiasis, hypercalcaemia, nephrocalcinosis and idiopathic hypercalciuria due to an increased absorption of calcium from the intestine (absorption hypercalciuria), the content of calcium in mineral water is recommended up to 150 mg/l. For better resorption and use of both ions, an advantageous ratio of calcium and magnesium is 2:1. Calcium resembles the basic building material for bones and teeth. It prevents colloid expansion, seals intercellular spaces, activates phagocytic chemotaxis and it has adstringent effect with catarrh inflammations. Its presence supports the diuretic effect of mineral water. Together with magnesium it decreases neuromuscular irritability and permeability of cellular membranes, affects a number of enzymatic reactions, increases immunity and it has an anti allergic effect. A higher content of magnesium in mineral water positively affects fatigue and depressions. Magnesium is the second most important intracellular cation after potassium, it is important for the work of the brain, heart and muscular cells. It belongs amongst the most powerful physiological vasodilators and it takes part in easing stress reactions.

Potassium

Its content in our mineral waters fluctuates between 5 mg/l in Ambrose spring to 87 mg/l in Cross spring. The recommended daily dose for adults is 4700 mg. The source of potassium is mainly fruit, vegetables, grains and milk products. Some interesting information was mentioned in a lecture by professor MUDr. J. Janda in Florence. Beer as a drink contains 20 - 110 mg of sodium, but 300 - 600 mg of potassium regardless if it is alcoholic or non-alcoholic beer.

While taking a drinking treatment we prefer to use cold mineral water for its stronger diuretic effect. In cold and damp weather, when having a cold and with patients, who recently underwent acute attack of pyelonephritis or cystitis, renal colic or suffer from symptoms of irritant urinary bladder, we prescribe lukewarm mineral water.

Patients suffering from digestive system illnesses and some metabolic disorders – through a drinking treatment we try to affect:

1. digestion, absorption, secretion, excretion and motility of the digestive tube along its whole length.
2. functions of the liver, gall bladder and gall tract including functions of the pancreas and affecting metabolism of saccharides, proteins and lipids.
3. we must also consider the secondary effect on the kidneys and urinary tract.



As a basic natural medicinal resource of drinking water - this indication has proved successful with mineral waters containing bicarbonate, sulfate and chloride anions bonded with cations of sodium, calcium and magnesium, hypotonic, isotonic and partially hypertonic. They are natural medicinal or acidulous waters.

Usually it is not mineral water of one composition (bicarbonate sodium like 'Bílinská kyselka'), but a combination of two or more mentioned anions and cations. In nomenclature, in the first place there is written anion and cation, which have a predominant effect on the mineral water. Other present components complete (strengthen or weaken) their effects.

We use drinking treatment for digestive system illnesses and some metabolic disorders in Mariánské Lázně:

I. MINERAL WATER WITH PREDOMINANT BICARBONATE ANION:

- a) bicarbonate-sulfate acidulous water, hypotonic (**Forest spring**)
- b) sodium-calcium bicarbonate-sulfate acidulous water, hypotonic (**Edward spring**)
- c) sodium-calcium bicarbonate-sulfate – chloride acidulous water, hypotonic (**Augustin spring**).

Alkaline mineral waters dissolve mucus, they have anti-inflammatory and astringent effects (Ca, Fe), neutralise hydrochloric acid, but they have two different effects on the secretion of acidulous stomach juice.

- a) It eases acidic stomach secretion when we serve warm mineral water, 1-2 cups of Forest spring or 1 cup of Edward spring or ½-1 cup of Augustin spring, 60 - 90 minutes before meals 3 times a day. Mineral water drunk in this way meets only a small amount of HCl in a stomach, which then neutralises and a predominant part of this consumed water goes in an unchanged condition to the duodenum, from where it affects, by reflex, the true moderation of acidic stomach secretion. Further intake of mineral water in between meals strengthens this effect. This effect is used for treating functional disorders and illnesses of stomach and upper parts of intestine with hyperacidity (ulcer disease of the stomach and duodenum in remission, conditions after stomach and duodenum surgeries, chronic gastritis with hyperacidity, functional stomach dyspepsia with hyperacidity).
- b) It stimulates acidic stomach secretion if the patients have been prescribed the same mineral water, but cold with a higher content of carbon dioxide, drunk 30 minutes before meals 3 times a day. Free carbon dioxide, present in acidulous water, and subsequently released carbon dioxide from bicarbonate during neutralisation of hydrochloric acid, causes hyperaemia of stomach mucosa and stimulates the secretion of acidic stomach juice. In that way it supports digestion and emptying of the stomach. The effect of mineral water received in this way is used with chronic gastritis and functional stomach disorders accompanied by hypoacidity.

Mineral water with predominant bicarbonate anion is suitable for treating some metabolic diseases using a drinking treatment for example diabetes mellitus type II, hyperuricemia and gout.

Diabetes mellitus II type

The basis of spa treatment of patients with diabetes remains the diet, adequate energy output, per oral antidiabetics, derivatives of insulin or insulin. The spa treatment is aimed at treating the accompanying and complicating symptoms of diabetes. The drinking treatment has a supporting meaning. We prescribe mineral waters with predominant bicarbonate anion, which after absorption, reduces acidity in tissues and increases sensitivity of peripheral tissues to insulin and depositing glycogen in the liver. This effect is increased by magnesium cation. Mineral water is prescribed in a small to larger amount according to the sodium concentration, divided into 3 - 4 individual doses. In Karlovy Vary, sodium bicarbonate-sulfate-chloride hot springs are used to treat diabetes and in Luhačovice, sodium bicarbonate-chloride acidulous water is used.

Gout, hyperuricemia with low urine pH, prevention of urate urolithiasis

With these problems, diet, allopurinol and the treatment of accompanying symptoms and also side illnesses have an essential significance. Mineral water with bicarbonate anion (Caroline, Rudolph, Balbin spring) is prescribed for drinking treatment. Induced polyuria causes an increased discharge of uric acid, but a parallel increase of diuresis lowers its concentration in urine. Urine pH higher than 6.5 increases dissolubility of uric acid 8 fold compared with pH 5. For alkalisation of urine we use springs with a predominant sodium bicarbonate (Forest, Edward and Augustin spring).

Concerning the excessive intake of sodium with cardiovascular diseases and hypertension, we most frequently combine both types of mineral water according to urine pH. Total daily amount (15 - 20 ml/kg per day) is divided into several individual doses (morning and afternoon – a spring with an alkalizing effect, before lunch and the evening meal - a spring with a diuretic effect).

Mariánské Lázně mineral waters are:

- a) sodium bicarbonate-sulfate acidulous water (**Forest spring**)
- b) sodium-calcium bicarbonate-sulfate acidulous water (**Edward spring**)
- c) sodium bicarbonate-sulfate-chloride acidulous water (**Augustin spring**)
- d) sodium-magnesium-calcium acidulous water (**Caroline spring**), suitable for diabetes with present complications (micro- and macroangiopathy).

II. MINERAL WATERS WITH PREDOMINANT SULFATE ANION

- a) sulfate-bicarbonate sodium acidulous waters, hypotonic to isotonic (**Cross spring**)
- b) sulfate-bicarbonate-chloride sodium acidulous waters, hypotonic to isotonic (**Alexandra and Ferdinand spring**)

Hypotonic and isotonic mineral waters dilute the stomach content by the intake volume and relatively quickly leave the stomach. Their main effects develop in further parts of the digestive tube and by influencing the function of the secondary digestive glands (liver, gall duct, gall bladder and pancreas). Only hypertonic mineral waters, which are missing in Mariánské Lázně, stay in the stomach much longer, because they need to be altered towards isotonia. Sulfates as bivalent cations are absorbed from the intestine slowly, according to their concentration.

Professor J. Benda has proved using **Karlovy Vary mineral water** (bicarbonate-sulfate-chloride sodium thermal water, hypotonic) an increase of gall secretion by liver cells (choleretic effect) as an answer to releasing or activation of secretin or pancreozymin type hormones in duodenum, by the intake of mineral water. After drinking warm mineral water and the increase of sulfates (1500 - 2000 mg/day), secretion of cholecystokinin occur together with an increase of gall bladder tonus, release of the sphincter of Oddi and emptying of the gall into the duodenum (cholecystokinetic effect) without any further unpleasant feelings (pressure in under the ribs, nausea, spasms). This effect starts 30 minutes after mineral water intake and it reaches its maximum around 90 minutes and a gradual declination takes approx. 2 hours (24). A sufficient amount of alkaline gall in the small intestine improves digestion, neutralises the remaining hydrochloric acid in the intestine content and increases the effect of products of fermentation secreted by small intestine glands and brought into the pancreatic juice.

More concentrated sulfate mineral waters has proven to have a laxative effect (about 3000 mg/day). This effect more is reliable in waters, which are in between hypo and isotonia like it is in Mariánské Lázně's mineral water in Cross and Ferdinand spring. A laxative dose is within the range of 750 ml, it means 2 cups in the morning and 1 before an evening meal, then it is altered, so we gain regular discharge of soft but not diarrhoea excreta. Sulfates contained in the mineral water stop resorption of osmotically bound water on colloids from the intestine content, this increases its volume and the pressure irritation of the intestine wall receptors causes quickening of peristalsis. Intestine content moved into the large intestine, where the excreta is finally formed, has then more water and more liquid consistency. Hydrogen sulfide, which is produced by sulfate reduction by bacterial intestine microflora, also contributes to the motility quickening. Hypotonic mineral waters with a lower concentration of sulfates have a more moderate and anti-inflammatory effect and positively affect intestine microflora.

Sulfate waters are suitable for drinking treatments curing liver, gall bladder, pancreas and intestine diseases. Warm mineral water is prescribed after having hepatitis, infectious mononucleosis, toxically damaged liver and chronic hepatitis, to the total daily amount of sulfates about 1000 - 1500 mg/day, divided into two to three individual doses. The treatment starts with a half amount, which is gradually increased according to the patient's tolerance. Another indication treated by drinking mineral water containing sulfate anions are diseases of the gall bladder and biliary tract (postcholecystectomy syndrome, chronic inflammation of the gall bladder, dyskinesia of biliary tract, cholelithiasis). Here it is also recommended to drink warm

mineral water of the total daily amount of sulfates of around 2000 mg, so choleric as well as colecystokinetic effects are reached. The overall daily amount is divided into individual doses, served 30 minutes before a meal. Approximately one half of the total amount before breakfast, ¼ before lunch and the last ¼ before an evening meal. It is recommended to drink warm mineral water in two doses 1 - 2 cups in the morning and before an evening meal with chronic pancreatopathy.

III. CHLORIDE ANIONS IN MINERAL WATER

For their composition, pure chloride mineral waters are not used here for drinking treatments. In mixed mineral waters they are usually bound with sodium or calcium cations and in accordance with bicarbonate anions they have either an easing or irritant effect depending on the temperature, carbon dioxide content and the time of served meals. Chloride anion in cold mineral water, rich in carbon dioxide, served 30 minutes before a meal, increases acidic stomach secretion. On the contrary, if the same mineral water is served warm, it has less carbon dioxide and it is drunk 60 minutes before a meal, it passes together with bicarbonates into the duodenum and from here it reflectorically subdues acidic stomach secretion. Both the above mentioned anions complement each other in affecting the secretion of stomach mucosa. Sodium chloride has a secretolytic effect, so it complements the mucolytic effect of bicarbonates. Calcium is used for its spasmolytic, anti-allergic and anti-inflammatory effects. It seals intercellular spaces, decreases the increased permeability of vascular walls in places of catarrhal inflammation, in doing so, it reduces inflammatory secretion, which becomes viscous and by subduing intestine peristaltic (through n. vagus) it contributes to excreta thickening with diarrhoea illnesses (therefore we sometimes observe an inclination to constipation with drinking treatment when using calcium-magnesium bicarbonate waters. Calcium chloride strengthens the effects of pancreatic juice during degradation of proteins and lipids in the small intestine.



While drinking mineral waters with digestive system illnesses, it is very important to follow a drinking treatment method. The recommended way to drink mineral water by sipping leads to a repeated irritation of taste receptors in the mouth and through the parasympathetic nervous endings to the initiation of secretion of digestive juices (hyperaemia of mouth mucosa induced by water temperature or present carbon dioxide, sour, salty, bitter taste). If more cups are prescribed, then we put a 10 minute break between drinks. Walking slowly whilst drinking the mineral water, makes it easier to drain from the stomach.

Mineral springs

Alexandra spring (Alexandřin pramen)



Sodium sulfate-bicarbonate-chloride acidulous water, cold, hypotonic, very strongly mineralised. It belongs to the outflow area of Ferdinand springs. It is prescribed to more sensitive patients with an inclination to diarrhoea excreta, where the stronger Cross or Ferdinand spring are not very well tolerated. The main indications are liver, gall bladder, pancreas and intestine illnesses. Regarding the total mineralisation and the content of sodium, even if it is hypotonic, this water is not suitable for a drinking treatment with kidney and urinary tract diseases, but it can be used as a completion of a drinking treatment using diuretic mineral water with chronic and repeated in-

flammations of the urinary bladder and with dysfunctional urination, to affect the intestine microflora and to induce regular defecation. Urine alkalisation with urate and cystin urolithiasis in combination with diuretic mineral water.



One spa cup (220 ml) contains approximately:

315 mg sodium, 365 mg bicarbonates, 360 mg sulfates, 130 mg chlorides

Ambrose springs (Ambrožovy prameny)

It has three outflows marked as I, II, III. They have very similar chemical and physical properties. They are pure, ferrous acidulous mineral waters, cold, hypotonic, slightly to medium min-



erally to medium mineralised. They have a diuretic effect, due to a low content of minerals, especially bicarbonate, they do not change urine pH, therefore they are suitable for drinking treatments with all types of urolithiasis including calcium phosphate, calcium carbonate or mixed. It is possible to also use them



for impact drinking with the same indications and counter-indications as it is stated in Caroline spring. They are suitable mineral waters to complement a drinking treatment with urine alkalisation using Forest or Edward spring to gain the desired increase of diuresis and does not burden the organism with an excessive intake of sodium. These acidulous mineral waters were once used to treat iron deficiency anaemia. Nowadays we use a number of effective preparations to treat anaemia, but these waters can be used to replenish the iron supply in the organism with subclinical anaemias, which often accompany chronic and repeated infections of various kinds (respiratory, urologic, gynaecologic). Bivalent iron (15 - 20 mg/l), which is contained in Ambrose springs, but also in other mineral springs in Mariánské Lázně, is better absorbed from fresh water drunk directly from a drinking fountain. It very easily oxidises after contact with the air to create trivalent compounds, which are harder to be absorbed and when the mineral water is bottled for a longer period, the compounds are released from mineral water in the form of rusty turbidity or sediments.

Antonin spring (Antonínův pramen)

It is publicly accessible in the gazebo at Antonin spring in Ůšovice, also called by older residents – ‘Antoníčkův’ spring. This calcium-magnesium bicarbonate, ferrous acidulous mineral water, cold, hypotonic, medium mineralised, with a content of carbon dioxide (2315 mg/l) is used by the locals as a tasty, refreshing drink. Up to now it is not used for treatments.

Augustin spring (Augustinův pramen)

It is one of the latest captured springs from the outflow structure of Ferdinand springs. It is a sodium bicarbonate-sulfate-chloride acidulous mineral water, cold, hypotonic, very strongly mineralised. The slightly stronger effect of bicarbonate anion is balanced by the presence of sulfate and chloride anions. The main field of influence of this mineral water is expected to be in a drinking treatment with digestive tract diseases. Regarding the overall mineralisation and the content of sodium, despite hypotony, it is not a suitable mineral water for a drinking treatment of kidney and urinary tract diseases. An exception can be its use for urine alkalisation with urolithiasis from uric acid or cystin (5 - 10 ml/kg/day divided into 3 amounts i.e. ½ - 1 spa cup), but sufficient diuresis has to be then induced by further drinking of diuretic water (Rudolph spring, Ambrose or Balbin spring). Its further use can be expected with uncomplicated diabetes, hyperuricemia with low urine pH and gout without symptoms of gout nephropathy.



One spa cup (220 ml) contains approximately:

410 mg sodium, 600 mg bicarbonates, 420 mg sulfates, 150 mg chlorides.

Balbin spring (Balbínův pramen)

A public drinking fountain is situated by the statues of the monarchs, at the beginning of a park in front of the building of hotel Bohemia. By its composition, the spring matches a medium mineralised magnesium-calcium-sodium bicarbonate ferrous acidulous mineral water, which is hypotonic and like the other springs in Mariánské Lázně it is cold. This hypotonic acidulous mineral water with a low content of sodium has a diuretic effect, it can suitably complement the choice of mineral waters with nephron-urological diseases, but in a larger amount (more than 600 ml/day) it is not suitable with phosphate lithiasis.



One spa cup (220 ml) contains approximately:

20 mg sodium, 18 mg magnesium, 22 mg calcium and 190 mg bicarbonates.

Edward spring (Edwardův pramen)

Together with Augustin spring it is one of the latest captured springs in the outflow area of Ferdinand springs. It is a sodium-calcium bicarbonate-sulfate acidulous mineral water, cold, hypotonic, very strongly mineralised with a high content of bivalent iron (48 mg/l). Also with this mineral water, its medicinal use, especially with stomach diseases with hypo as well as hyper acidity, inflammatory diseases of liver, gall bladder, pancreas and intestine, can be presumed. Despite hypotony, it is not a suitable mineral water for treating kidney and urinary tract diseases (it is very strongly mineralised with a high content of sodium and bicarbonate). Based on analysis, we can presume its use for alkalinisation of urine with cystin and urate lithiasis, for drinking treatment with gout combined with diuretic mineral water and with uncomplicated diabetes.



One spa cup (220 ml) contains approximately:

222 mg sodium, 77 mg calcium, 40 mg magnesium, 525 mg bicarbonates, 290 mg sulfates.

Ferdinand spring (Ferdinandův pramen)

According to analysis, it is classified as a sodium sulfate-bicarbonate-chloride acidulous mineral water, cold, hypotonic to isotonic, very strongly mineralised. And like Cross spring, it has choleretic effects (small amounts), cholecystokinetic effects (medium amount) as well as laxative effects (medium to large amount) and the same indication specification as Cross spring. It is prescribed for drinking treatment for digestive system diseases in full extent, spastic as well as atonic constipation. It has a reliable laxative effect. Combined with diuretic mineral water, it is prescribed for chronic and repeated inflammations of urine bladder, dysfunctional urination and repeated gynaecologic inflammations to affluence intestine microflora and induce regular defecation. Formerly a laxative salt, which was used as an additive to water

for a home drinking cure, was produced from Ferdinand spring by evaporation. This year (2018) its production has been renewed.



One spa cup (220 ml) contains approximately:

455 mg sodium, 530 mg bicarbonates, 500 mg sulfates, 170 mg chlorides.



Joseph spring (Josefův pramen)

It is one of the youngest springs in Mariánské Lázně. It is a sodium-magnesium-calcium bicarbonate-sulfate ferrous acidulous mineral water, cold, hypotonic, medium mineralised with a high content of free carbon dioxide (3795 mg/kg), with a refreshing taste. It can be used for drinking treatment for urolithiasis of unknown composition, to support the appetite with stomach hypoacidity and as a refreshing drink.



One spa cup (220 ml) contains approximately:

40 mg sodium, 15 mg calcium, 10 mg magnesium and 150 mg bicarbonates, 50 mg sulfates.

Caroline spring (Karolinin pramen)



It is a sodium, magnesium, calcium bicarbonate acidulous mineral water, cold, hypotonic, medium to strongly mineralised. It has a similar impact on the kidneys and urinary tract as Rudolph spring. It has a diuretic effect, urine pH moves slightly towards neutral, but it does not change it significantly, therefore it is suitable with all nephron-urolithic diseases, especially hypercalciuria for

a lower amount of calcium than is in Rudolph spring, with oxalate urolithiasis for it better ratio of magnesium and calcium. It is prescribed in the amount of ml/kg/day divided into 3 - 4 doses (in the morning on an empty stomach, before lunch and the evening meal alternatively before sleep). Do not exceed the amount of 2 litres per day. While drinking the amount of 0.5l three times a day, the daily intake of calcium (850 mg/day) by increases by another 15% and the daily intake of magnesium (400 mg/day) by 40%. Low diuresis and an excessive concentration of calcium in the urine belong to the main risk factors of formation and recidivity of calcium stones, which could be affected by a drinking treatment. Through higher water diuresis during a drinking treatment using Caroline spring causes an increased excretion of magnesium as well as decreasing of calcium/magnesium quotient and in this way it decreases the risk of precipitation of oxalates (16). In indicated cases, it is possible to use it also for so called impact drinking, which helps the spontaneous discharge of small stones (up to 6 - 8 mm), follow-up treatment after crushing the stones or at random with microscopic haematuria and crystalluria. It is prescribed in the amount of 30 ml/kg/day, the patient should drink 2/3 of the amount (maximum 2 litres) in the morning during 2 - 3 hours, the last third in the late afternoon. After impact drinking, he/she has prescribed various sport activities (exercise on a ball, jumping on the stair, skipping, exercise in a pool, Nordic walking, walking on an uneven terrain) during the whole day. During the day, the patient urinates through a sieve to catch discharged stones for chemical analysis. From a practical point of view, we prescribe impact

drinking 1 - 2x a week, during the first half of the week. Counter-indications: acute febrile illness, with seniors and pre-school children, congenital kidney and urine tract anomalies, vesico-uterine reflux of higher than the 1st level, advanced pyelonephritis and glomerulonephritis, cast stone lithiasis, urolithiasis with a possible obstruction (signs of congestion), ren migrans (movable kidney), constriction of ureter and urethra, advanced stadia of a prostate gland adenoma, functional disorders when urinating, advanced sclerosis of the urine bladder neck, use of antidiuretic hormone (Minirin, Adiuretin), hypertension, all levels of cardiac decompensation, glaucoma, varix veins of oesophagus, stomach and duodenum ulcer disease.

Another indicationa for drinking treatment of Caroline spring are diabetes mellitus II including diabetic nephropathy, gout also with nephropathy.



One spa cup (220 ml) contains approximately:

40 mg sodium, 30 mg calcium, 20 mg magnesium and 210 mg bicarbonates.



Cross spring (Křížový pramen)

It is one of the longest used springs for drinking treatments in Mariánské Lázně. It has the highest total mineralisation and the highest concentration of sulfates. Based on chemical analysis and physical properties, it is defined as a sodium sulfate-bicarbonate ferrous acidulous mineral water, cold, hypotonic to isotonic, very strongly mineralised. It has choleric, cholecystokinetic as well as laxative effects. With gall bladder diseases we prescribe warm water 1 cup 2 - 3x a day. With symptoms of irritated colon and ulcerous colitis we prescribe warm water, in the morning on an empty stomach and before an evening meal

1 cup of Rudolph spring, before lunch ½ - 1 cup

of Cross spring with further moderation of the amount according to the type of excreta. With spastic constipation we recommend to drink warm water with a lower content of carbon dioxide slowly sipping 3 x 1 cup of Cross spring half an hour before a meal. On the contrary, with atonic type constipation, patients drink cold water or alternating one cup of warm and one of cold water. Usually we start with 1 - 2 cups in the morning and one cup before an evening meal. With both types of constipation, we gradually increase the amount of water until we achieve an excreta of soft consistency. Not watery diarrhoea. Combined with diuretic mineral water we use it with chronic and repeated gynaecologic inflammations to affect the intestine microflora and to induce regular defecation.



One spa cup (220 ml) contains approximately:

590 mg sodium, 690 mg sulfates, 625 mg bicarbonates



Forest spring (Lesní pramen)

For years it was prescribed for giving the strongest alkalogenous effect from Mariánské Lázně springs. It is a sodium bicarbonate-sulfate acidulous mineral water, cold, hypotonic, strongly mineralised. As hypotonic water (135 kPa) it also has a diuretic effect. Due to the predominant effect of sodium bicarbonate it is indicated for drinking treatment with stomach and duodenum illnesses.

With functional dyspeptic problems and with chronic gastritis with decreased secretion of acidulous stomach juice – hypoacidity, we prescribe 1 - 2 cups of cold mineral water 3x day drunk 30 minutes before a meal. Free carbon dioxide contained in cold mineral water stimulates stomach motility and secretion of acidulous stomach juice, which is strengthened by secondarily released carbon dioxide created by the reaction of hydrochloric acid with sodium bicarbonate. With hyperacidity conditions we prescribe warm Forest spring. We recommend drinking it after stirring (releasing carbon dioxide) approximately 60 - 90 minutes before a meal. Mineral water neutralises a small amount of hydrochloric acid and almost unchanged, it reaches the duodenum. Here it alkalises its content and reflectorically subdues acidulous stomach secretion. Bicarbonate is resorbed especially from duodenum, after resorption metabolic alkalosis does not happen, however it moderates acidity in peripheral tissues and by this it improves their sensitivity to insulin and it supports depositing of glycogen in the liver. Therefore, Forest spring is also prescribed for drinking treatment with patients with diabetes mellitus II type. A part of resorbed bicarbonate is discharged by the liver cells into the bile, a part is secreted in the ileum in exchange for acidic chloride anions, but most of it is then discharged by the kidneys to urine, together with sodium and potassium cations, moving pH toward neutral up to alkaline values (pH 6.5 - 7.2). We use this effect to alkalise the urine with urate and cystin urolithiasis. We alkalise urine only if there is no urine infection and the pH does not exceed 7.2, due to the threat of phosphate crystallisation! The usual amount of Forest spring is 15 - 20 ml/kg/day divided into 3 - 4 individual doses up to the maximum amount of 2 litres, if there are no counter-indications.



One spa cup (220 ml) contains approximately:
120 mg sodium, 300 mg bicarbonates and 110 mg sulphates.

Spring St. Maria Eva (pramen sv. Marie Eva)

The publicly accessible drinking fountain of Maria spring is situated in a gazebo in front of hotel Esplanade. It is a pure acidulous water, cold, hypotonic, slightly mineralised, richly gassed with carbon dioxide. It is a refreshing drink containing sodium, formerly bottled under the brand name Aqua Maria.

Rudolph spring (Rudolfův pramen)

It is a founder the fame of Mariánské Lázně as a urologic spa. It is defined as a calcium-magnesium bicarbonate acidulous water, cold, hypotonic, strongly mineralised. It has the strongest diuretic effect of Mariánské Lázně's springs, which is conditioned by the induced water diuresis by drinking hypotonic mineral water with a low content of sodium and it is strengthened by the presence of bivalent calcium and magnesium cations. Through this mechanism of repeated leaching of pathological parts of the urine, such as the peeled off epithelium, leucocytes, erythrocytes and bacteria during inflammations of the upper and lower urinary tract, urine sand, crystals of stone forming matter and small stones. The induced polyuria also affects the peristalsis of ureters and the collecting function of the urinary bladder. During escalated diuresis, induced by hypotonic mineral water with a low content of sodium, the mechanical pressure of the urine flow affects the ureter walls and provokes their peristalsis, accelerates the course and size of urine boluses. When diuretic response culminates (approx. after 90 minutes), individual boluses fusion occurs and ureters are temporarily filled throughout their whole length. With fading acute diuretic response, urination in peristaltic waves is restored again. We use this effect of a drinking treatment mainly with scheduling a larger amount (2 - 3x a day) with patients with urolithiasis, then with dysfunctional peristalsis of the urinary tract due to chronic and relapsing urinary inflammations and conditions after surgery. Quicker filling of the urinary bladder increases its irritation and decreases capacity, which is used to the advantage of patients with an inclination to hypotony of the urinary bladder and postponing of miction. We divide the total daily amount of cold mineral water into two larger individual amounts (in the morning on an empty stomach and in the afternoon at around 3 - 4 p.m.).

On the contrary, we prescribe the daily amount of warm or at least lukewarm mineral water divided into smaller amounts for patients, who have an irritated urinary bladder with a smaller capacity. It means, that the total daily amount of mineral water is split into 6 - 8 smaller amounts to achieve gradual filling of the urinary bladder and increase its capacity. For patients with nocturia, enuresis nocturna and vesicoureteral reflux we recommend to drink the mineral water at the latest around 4 to 5 p.m.

While drinking mineral water rich in calcium, such as Rudolph spring, absorption of calcium increases and the excess is then discharged in the urine. Calcium from the small intestine is absorbed by active and passive transport. Active transport is controlled by hormones and it is limited. Passive transport does not depend on vitamin D and it is applied only during high calcium supply. With normocalciuria, an increased discharge of calcium is compensated by increased diuresis and an increased discharge of magnesium, but with calcium urolithiasis, especially with idiopathic hypercalciuria of absorption type, a drinking treatment using Rudolph spring means a further increase of the already pathologic values of calciuria! In these cases, we prescribe a drinking treatment using slightly mineralised pure acidulous mineral waters (Ambrose spring) or mineral water with a low calcium content (Caroline, Balbin spring).



One spa cup (220 ml) contains approximately:

60 mg calcium, 30 mg magnesium, 20 mg sodium and 350 mg bicarbonates.

Drinking treatment using Rudolph spring is indicated:

- ≈ with chronic and repeated inflammations of the upper and lower urinary tract (pyelonephritis, cystitis)
- ≈ with conditions after surgery of the urinary system
- ≈ with urolithiasis without present hypercalciuria
- ≈ with oxalate urolithiasis, we can combine the drinking treatment using Caroline spring (30 minutes before a meal 3x a day) with Rudolph spring served in between main meals (approx. 1 - 1.5 hours after breakfast and lunch). The calcium received with mineral water reacts in the intestine with soluble oxalate salts producing poorly soluble compounds, from which oxalate cannot be absorbed and it is discharged with excreta. Oxaluria decreases, discharge of magnesium by urine increases, which inhibits the crystallisation of oxalates.
- ≈ with glomerulonephritis, nephrotic syndrome and after kidney transplantation. Usually of a daily amount 5 - 10 - 15 ml/kg/day in three individual doses to treat the depletion of calcium and magnesium due to immunosuppressive syndrome.
- ≈ with secondary hyperoxaluria after intestine resection or malabsorption syndrome during regional enteritis, celiac disease, chronic pancreatitis to prevent oxalate urolithiasis creation. Unabsorbed calcium bonds free oxalates, decreases an increased permeability of the intestine wall and through these two mechanisms it prevents their absorption.
- ≈ supportive treatment of locomotor system diseases with osteoporosis, where a high content of calcium, a bond to bicarbonate, present carbon dioxide and calcium – magnesium ratio in mineral water 2:1 positively affect their use and resorption from the intestine.
- ≈ supportive treatment for muscle spasms.
- ≈ period of increased demand of calcium and magnesium, such as during growth acceleration, osteoporosis, pregnancy, lactation, with sportsmen and also seniors.

Due to a high content of calcium, a higher intake of Rudolph mineral water is unsuitable with calcium urolithiasis and hypercalciuria. Absolutely counter-indicated is with absorption hypercalciuria (pathological discharge of calcium into the urine during its increased absorption from the intestine).

Central spring (Ústřední pramen)

It is a pure ferrous acidulous mineral water, slightly mineralised with a content of bivalent iron about 20 mg/l and carbon dioxide 2400 mg/l suitable as a refreshing drink. It has diuretic effect and in a larger amount it is not suitable with ulcer diseases of the stomach and conditions with hyperacidity. It is accessible from a hotel drinking fountain in the corridor in front of the pool with carbonated gas from Maria spring.

Public drinking fountain in the 'area of health' - Prelat spring (Prelátův pramen)

A calcium-magnesium bicarbonate (Ca 121, Mg 60.2 mg/l) ferrous (Fe 39.4 mg/l) acidulous mineral water (2570 mg CO₂/kg), cold, hypotonic (61 kPa), medium mineralised

(solid residue after evaporation 845 mg/l) is brought to this drinking fountain. It comes from the well "Sonda V" and it is regularly checked. It is used especially by children while they play in the 'area of health' as a refreshing drink. It is not used for drinking treatments.

As is written by a doyen of spa nephrology prim. MUDr. Vladimír Křížek, CSc. in his book about drinking treatments and completed by the authors according to the currently accessible mineral water springs for drinking treatment.

REMEMBER, in Mariánské Lázně:

The highest resorption availability of calcium:	Rudolph spring
The lowest concentration of calcium:	St. Maria Eva, Central, Ambrose spring
The highest concentration of sodium:	Cross, Ferdinand, Augustin, Alexandra, Edward spring
The lowest concentration of sodium:	Central, St. Maria Eva, Antonin, Ambrose, Rudolph spring
The highest resorption availability of magnesium:	Rudolph, Caroline spring
The highest concentration of bicarbonate:	Cross, Augustin, Ferdinand, Edward, Alexandra, Forest spring
The highest concentration of sulfates:	Cross, Ferdinand, Augustin, Alexandra, Edward spring
For refreshment, you can drink from Mariánské Lázně's springs, which have a low content of salts, for example	Ambrose, Antonin, Balbin, Joseph, St. Maria Eva and Central spring

Mineral waters in Mariánské Lázně widen the choice of drinks, but even these mineral waters should not be the only source of liquids in your diet.

Drinking treatments can help, but also harm. Like medicines, they have indications and counter-indications. Some mineral waters contain grams of sodium and they are unsuitable for people with cardiac diseases, hypotony, liver cirrhosis, kidney failure, small children and seniors!

Usually the same mineral waters also contain grams of sodium bicarbonate. Alkaline pH of urine and present infection of the urinary tract can lead to forming and clustering phosphate salts in the form of phosphate stones or their adhesion and growth of the phosphate skin around the already existing urate or oxalate stones.

Grams of sulfates contained in some of Mariánské Lázně's mineral waters decrease the resorption availability of some ions and complicate their use in a larger drinking dose. With an unsuitable amount, watery diarrhoea with a decrease of urine discharge, dehydration and disintegration of the inner environment can occur.

TRUST YOUR DOCTOR

and keep the choice and dosage of mineral water according to the prescription.

Used literature:

- JANDA J., Salt intake and blood pressure, 7. Evropský kongres EPA-UNEPSA, Florencie, 5/2015
- ŠPIŠÁK L., RUŠAVÝ Z., a kol., Klinická balneologie, Nakladatelství Karolinum, Praha, 2010,
- JANDOVÁ D., Balneologie, Praha, 2009
- KOŽÍŠEK F., JELIGOVÁ H., Zdravotní význam sodíku ve vodách, SZÚ Praha, 1/2002
- ZÁKON č.164/2001 Sb. O přírodních léčivých zdrojích, zdrojích přírodních minerálních vod, přírodních léčebných lázních a lázeňských místech (lázeňský zákon), Praha,2001
- VYHLÁŠKA č.423/2001, kterou se stanoví způsob a hodnocení přírodních léčivých zdrojů a zdrojů přírodních minerálních vod...(vyhláška o zdrojích a lázních), MZ, Praha, 2001
- BENDA J., SADÍLEK L., Přírodní hydrouhličitanové minerální vody v balneoterapii, Rehabilitace a fyzikální lékařství, č. 4/2000
- BENDA J., Pitná léčba v lázních (mechanismus působení přírodních minerálních vod), Rehabilitace a fyzikální lékařství, č. 3/1996
- SCHÜCK O., TESAŘ V., TEPLAN V. a kol., Klinická nefrologie, Medprint, Praha, 1995
- KŘÍŽEK V., SADÍLEK L., Pitná léčba v urologii a nefrologii, VÚB Mariánské Lázně, Balneologické listy 15, 1987
- KŘÍŽEK V., SADÍLEK L., Některé důsledky velkého příjmu tekutin a velké diurézy, VÚB Mariánské Lázně, 1986
- KŘÍŽEK V., Lázeňská léčba chorob ledvin a močových cest, VÚB Mariánské Lázně 1986
- KŘÍŽEK V., SADÍLEK L., Pitná léčba Baldovskou kyselkou, VÚB Mariánské Lázně,1984
- KŘÍŽEK V., SADÍLEK L., Lázeňská léčba urolitiázy, VÚB Mariánské Lázně, 1984
- VENDL L., Infekce močových cest a fosfátová urolitiáza v dětském věku, Čs. Pediatrie č.8,1982
- KŘÍŽEK V. a kol., Karolinin pramen. Návrh na využití Karolinina pramene a studie pitné léčby, VÚB Mariánské Lázně, 1981
- KŘÍŽEK V., SADÍLEK L., Vliv pitné léčby karlovarským Mlýnským pramenem, VÚB Mariánské Lázně, 1981.
- KŘÍŽEK V., SADÍLEK L., Pitná léčba ve vztahu k hypercalciurii. Pokus o medikamentózní ovlivnění Hydrochlorothiazidem, VÚB Mariánské Lázně, 1980
- KŘÍŽEK V., SADÍLEK L., Metabolická problematika pitné léčby, ovlivnění metabolismu kalcia pitnou léčbou IV, VÚB Mariánské Lázně, duben 1980.
- KŘÍŽEK V., KUŽEL K., Klinické sledování lázeňské léčby urolitiázy, VÚB Mariánské Lázně, 1977.
- SADÍLEK L., KŘÍŽEK V., Vztahy mezi příjmem kalcia potravou a výdejem močí, VÚB Mariánské Lázně, 1977
- KŘÍŽEK V., Studie o minerální vodě Rudolfe, Balneologické listy, ročník IV., č. 12, VÚB Mariánské Lázně, 1976
- KOHLÍČEK J., Lázeňská léčba urologických nemocí, Balnea Praha, 1971
- BENDA J., Vliv karlovarské minerální vody na tvoření a vylučování žluče. Kandidátská disertační práce, Lázeňská sanatoria Imperial Karlovy Vary, Výzkumný ústav výživy lidu Praha, 1964
- KUŽEL K., Diuretické účinky některých minerálních vod. Lék. Listy 7:206,1952

Mineral springs in Mariánské Lázně for mineral water drinking cure

Complete analysis:		2013	2013	2014	2013
Check analysis:		2017	2017		2016
Spring:		Alexandra	Ambrose	Augustine	Balbin
		BJ 12	I,II,III	HV 43	
pH:		6,18	5,65	6,09	5,9
osmotic pressure		384 kPa	25 kPa	456 kPa	64 kPa
Cations mg/l					
Sodium	Na	1430	55,3	1880	93,4
Magnesium	Mg	72	24	131	79,4
Calcium	Ca	91,8	57,7	219	102
Iron	Fe	18,9	15,6	27,7	21,7
Anions mg/l					
Chlorides	Cl	601	57,8	704	30,4
Sulfates	SO ₄	1630	64,4	1910	124
Bicarbonates	HCO ₃	1654	288	2680	854
Total mineralisation:	mg/l	5497	469-563,55	7735	1304,96
free carbon dioxide	CO ₂ mg/kg	3750	2580	2325	2875
Classification:		sulfate - bicarbonate-sodium chloride ferrous acidic water cold, hypotonic, very strongly mineralised	pure ferrous acidulous water cold, hypotonic, slightly-medium mineralised	bicarbonate-sulfate - chloride sodium ferrous cold, hypotonic, very strongly mineralised	bicarbonate-magnesium - calcium - sodium ferrous acidulous cold, hypotonic, medium mineralised
Indications:		Liver, gall bladder, pancreas and intestines diseases urine alkalization	Urolithiasis of unknown composition including infectious urolithiasis, hyperuricemia, gout with nephropathy	Digestive system diseases, constipation	Oxalate urolithiasis, idiopathic hypercalciuria, DM II. with micro- and macro-angiopathy obesity
Location of the public drinking place:		Alexandra spring gazebo in a park under hotel Butterfly	In front of hotel Centrální Lázně a group of Ambrose springs I,II,III	In front of hotel Cristal	by the statues of the monarchs in the park in front of hotel Bohemia
Hotel drinking places:		Falkensteiner GH	Connecting corridor between hotels Centrální Lázně and Nové Lázně	In the hotels Cristal and San Remo	Hotels Centrální Lázně, Hvězda, Vltava